Computer Vision

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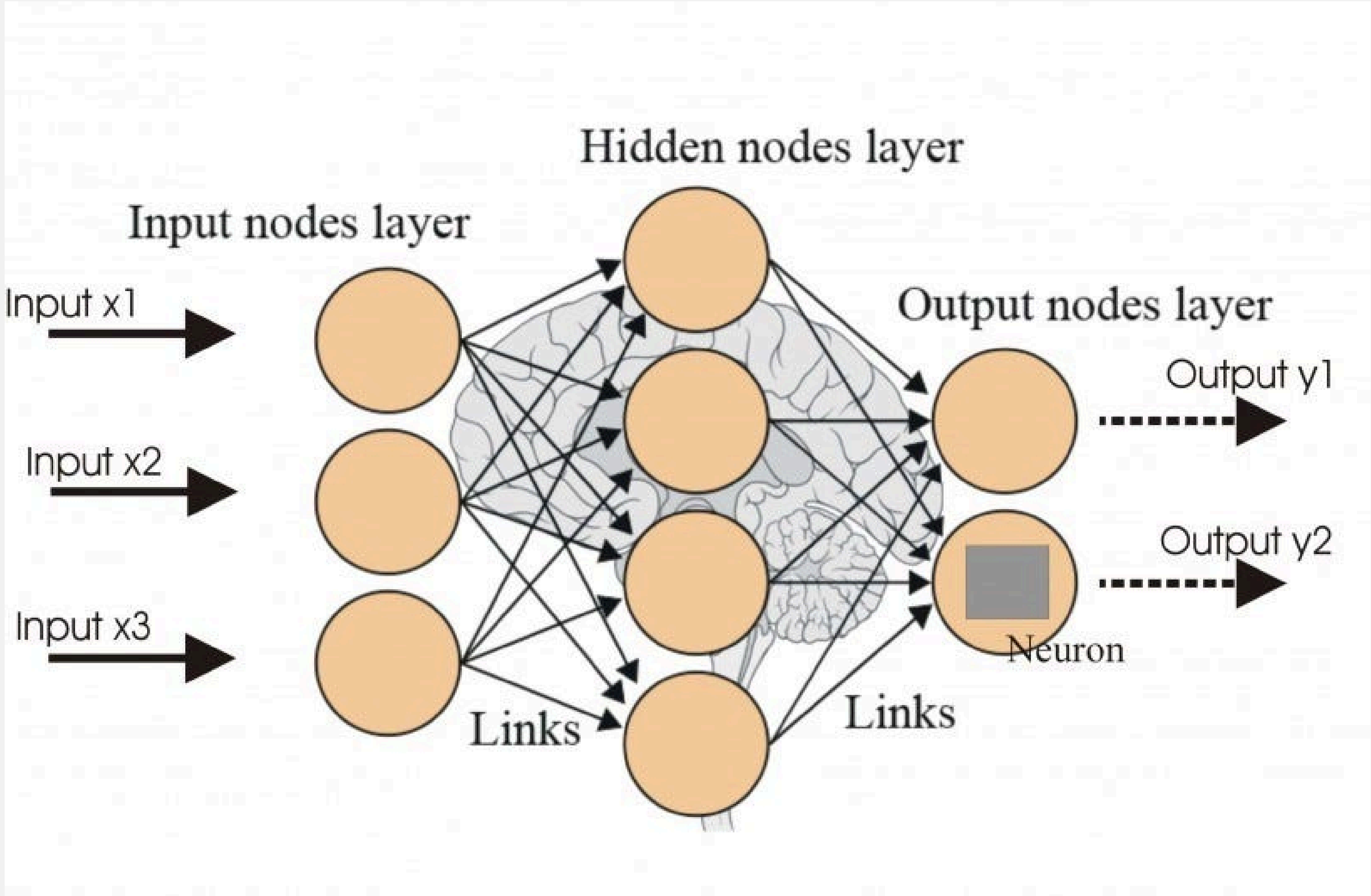
ARTIFICIAL NEURAL NETWORKS (ANN)



What is an ANN?

- Inspired by the biological brain (neurons & synapses).
- Learn complex patterns from data.

Components:

- Input Layer: Receives raw features.
- Hidden Layers: Process information (depth = "deep learning").
- Output Layer: Produces prediction.

- Strengths:
  - Works on tabular data (spreadsheets, CSV).
  - Solves non-linear problems (XOR, classification).
- Limitations (The "Curse of Dimensionality"):
  - Ignores spatial structure (treats pixels as independent).
  - Needs flattening of images →→ loses neighborhood info.
  - Many parameters →→ computationally heavy for images.

CONVOLUTIONAL NEURAL NETWORKS (CNN)

Why CNN for Images?

- Problem with ANN on a 32x32 image:

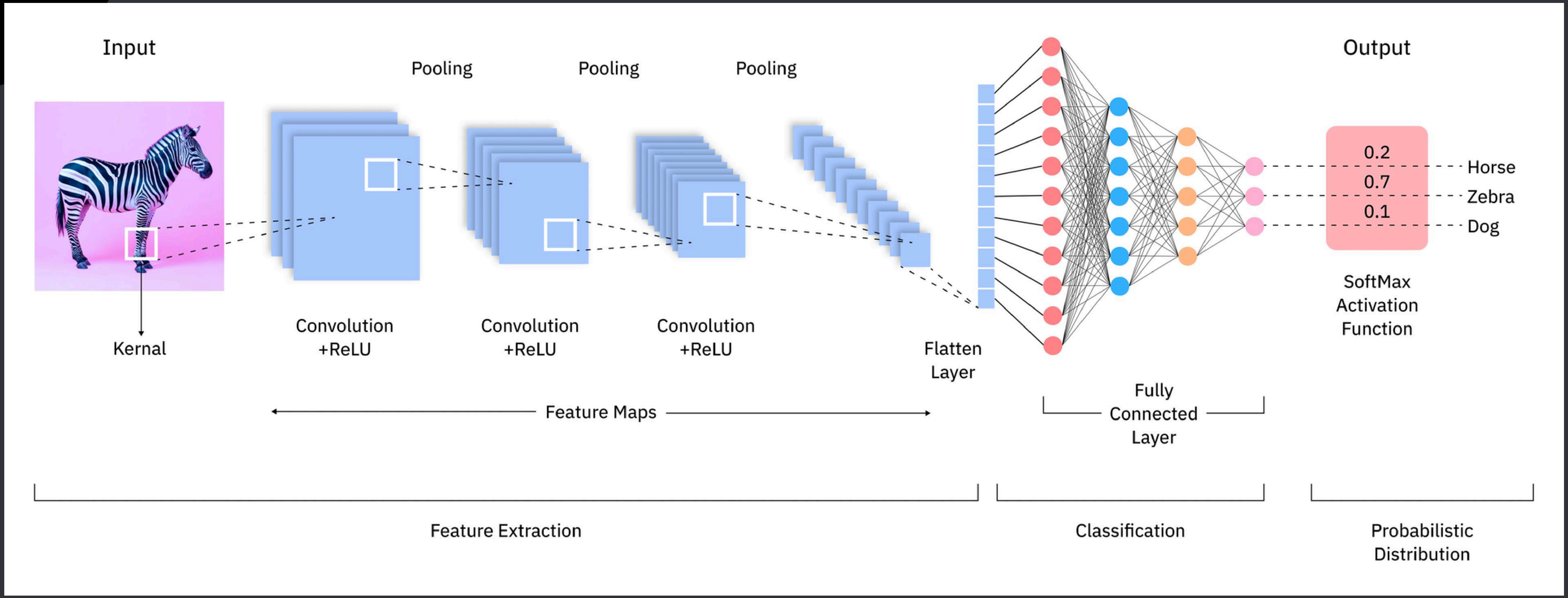
1024 pixels×1000 neurons = 1M weights (easy overfitting).

CNN Solution: Use spatial priors.

- Local Connectivity: Look at small patches.
- Parameter Sharing: Same filter across entire image.

Main Components

- Convolution Layer
- Activation Function (ReLU)
- Pooling Layer
- Fully Connected Layer



Input Image
↓
Convolution
↓
Feature Maps
↓
ReLU
↓
Pooling
↓
Flatten
↓
Dense Layer
↓
Prediction

CNN Working

CNN is designed specifically for image processing.

Convolution Operation

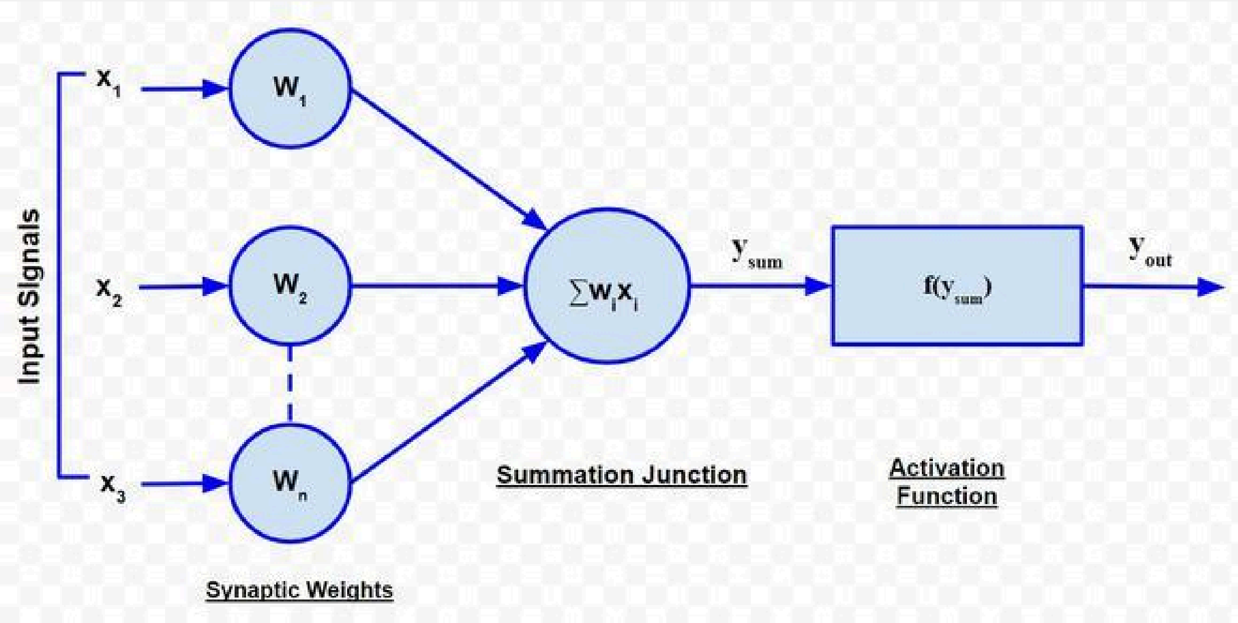
What does CNN learn?

Early Layers:
- Edges
- Corners

Middle Layers:
- Shapes
- Patterns

Deep Layers:
- Objects
- Faces

Benefits
- Parameter Sharing
- Feature Extraction
- Translation Invariance

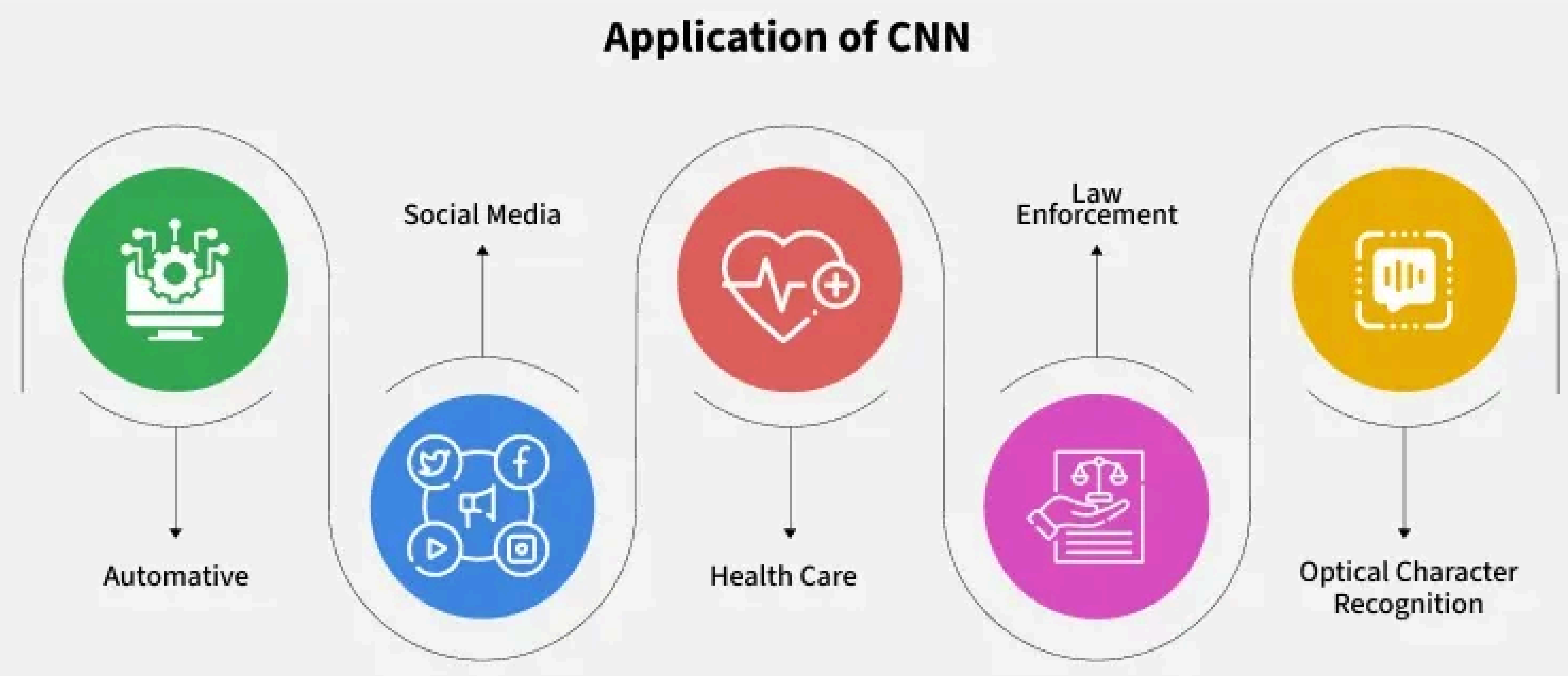


LIMITATIONS

- Excellent for images
- Weak at understanding long sequences
- Limited context awareness

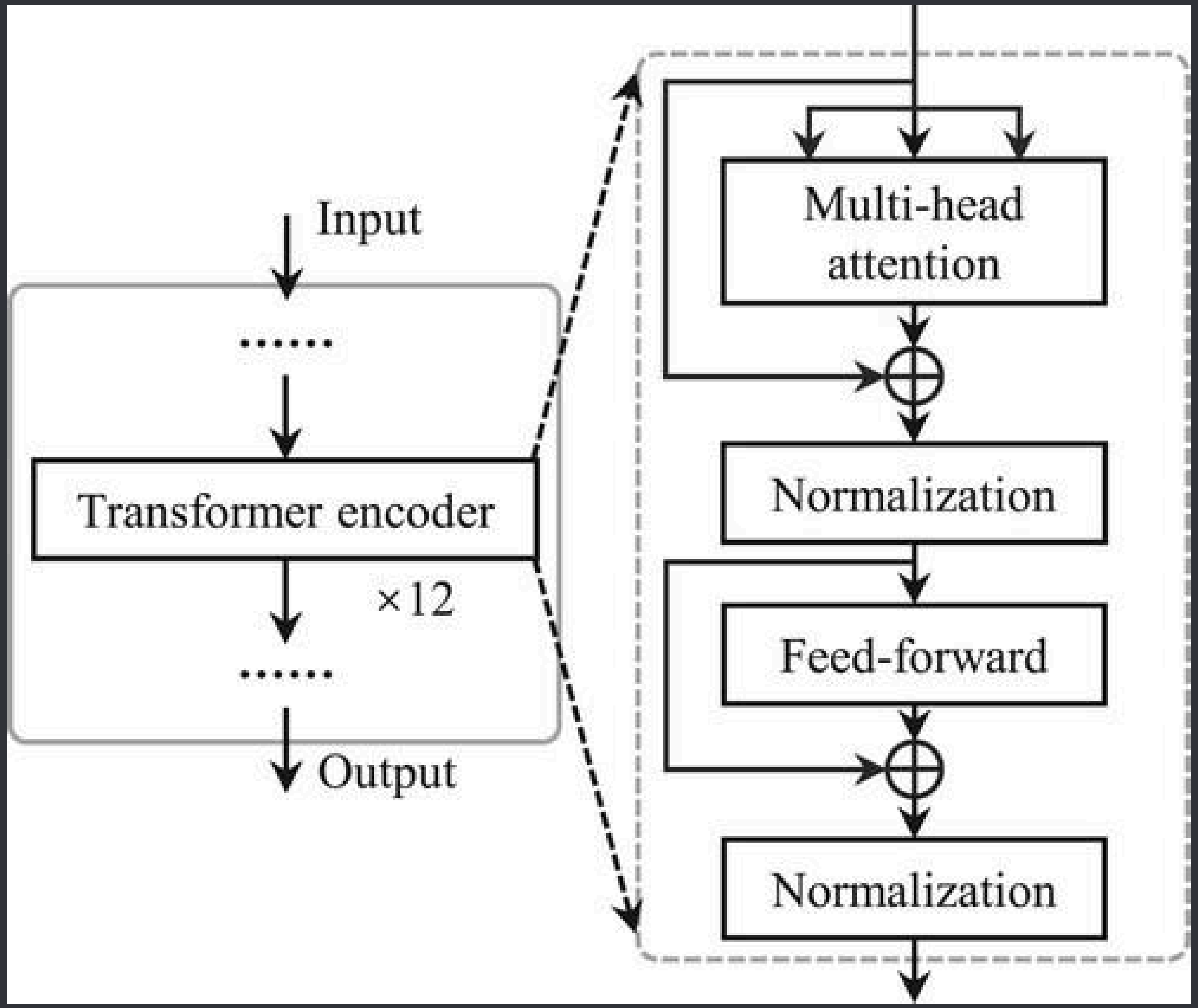
APPLICATION

- Face Recognition
- Medical Imaging
- Object Detection
- Self Driving Cars
- Security Systems



TRANSFORMER ENCODER

Attention Is All You Need

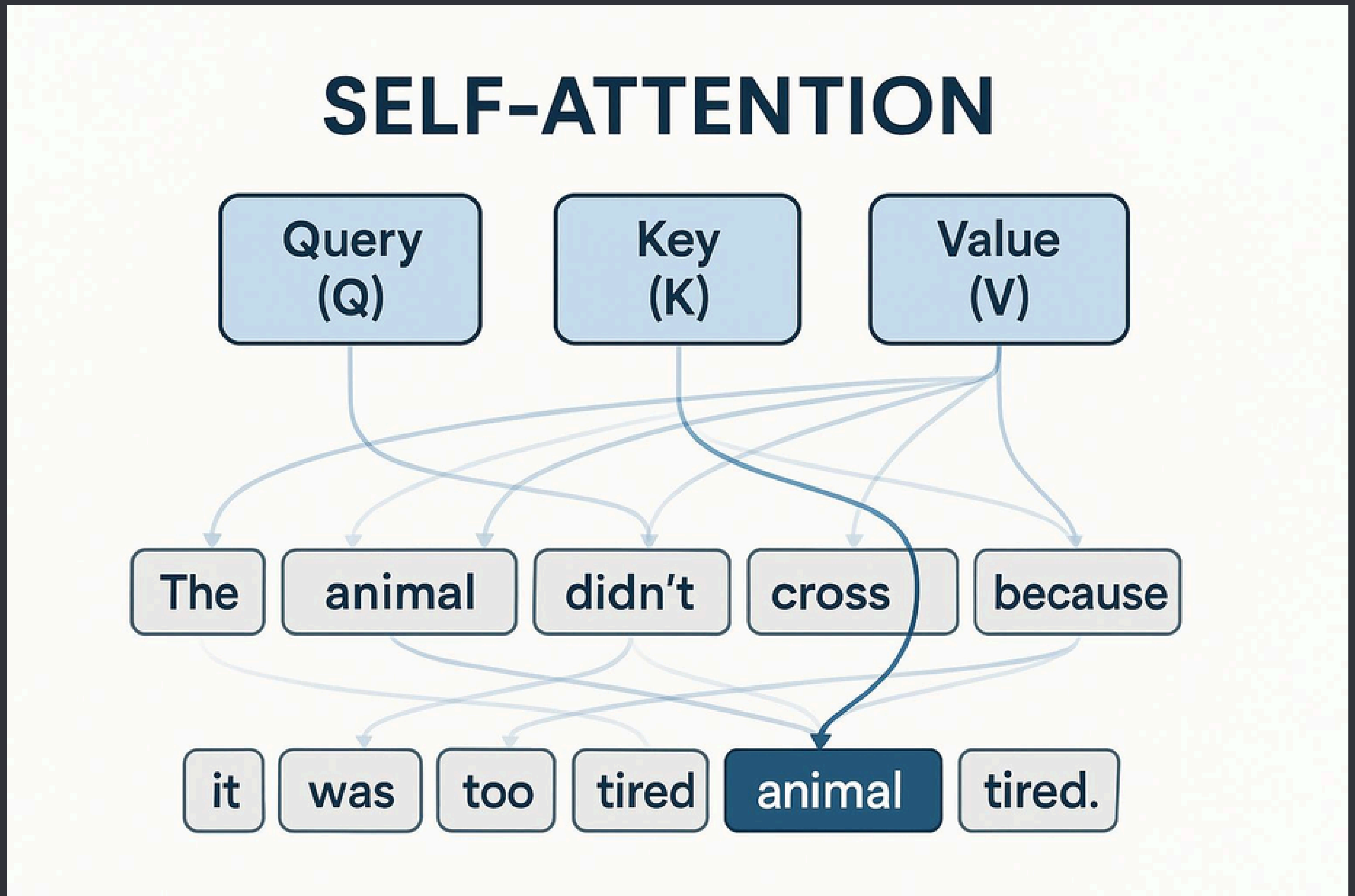


<u>Main Components:</u>

- Input Embeddings
- Positional Encoding
- Multi-Head Attention
- Feed Forward Network
- Layer Normalization

Self-Attention Mechanism

- Goal: For each word, look at ALL other words in the sentence and decide how much to "attend" to them.
- Example Sentence: "The animal didn't cross the street because it was too tired."
  - "it" attends strongly to "animal" (not "street").



Three Vectors

- Query (Q)
- Key (K)
- Value (V)

Comparison

Feature	**ANN**	**CNN**	**Transformer**
Data Type	Tabular	Images	Text/Sequence
Feature Extraction	Manual	Automatic	Automatic
Context Understanding	Low	Medium	High
Parallel Processing	Yes	Yes	Yes
Modern NLP	No	No	Yes

Thank You